Yuwei (Johnny) Meng | CSC2412

MEMORIZATION IN DP FINE-TUNED LLMS

An Empirical Study of Differential Privacy

Introduction & Background

Since the invention of the transformer (Vaswani et al., 2017), LLMs have been widely adopted for NLP tasks. However, training requires large text corpora that may contain sensitive information. Differential privacy (DP) offers a formal guarantee:

$$P\left(M\left(X\right) \in S\right) \leqslant e^{\varepsilon} P\left(M\left(X'\right) \in S\right) + \delta$$

In practice, DP-SGD adapts gradient descent by clipping gradients and adding Gaussian noise. Carlini et al. (2019) showed that standard training can cause models to memorize fine-grained information (e.g., SSNs), and introduced canary insertion — injecting formatted secret strings to measure memorization. This study applies canary insertion to a privacy-sensitive mental health dataset to assess how effectively DP-SGD prevents memorization.

Research Question

At what privacy budget ε does DP-SGD effectively prevent memorization of sensitive sequences in fine-tuned LLMs, and what is the accuracy cost?

Canary

“My patient ID is XXXXXX”

Log-Perplexity

$$l_{\theta}\left(x_1, \ldots, x_n\right) = -\frac{1}{n}\sum_{i=1}^{n} \log_2 P\left(x_i \,|\, f_{\theta}(x_1, \ldots, x_{i-1}\right)$$

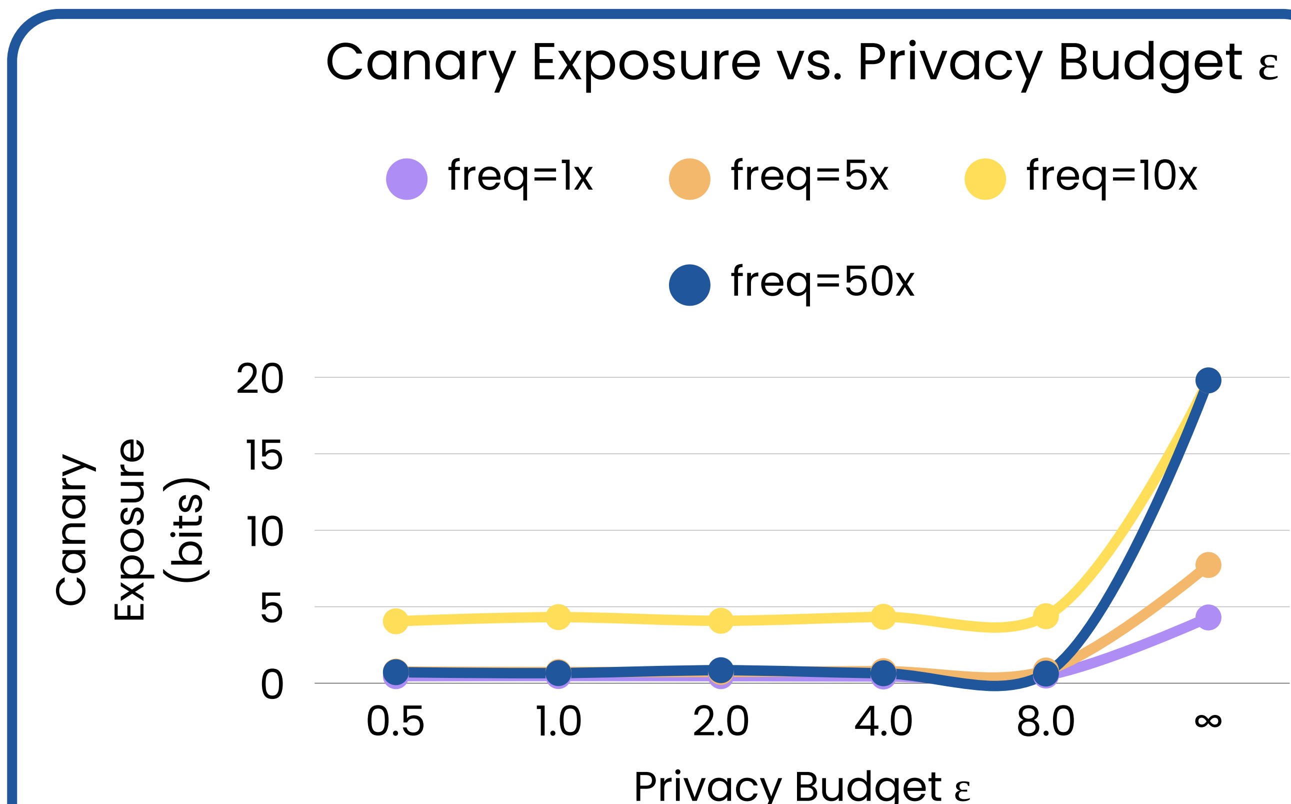
Canary Exposure

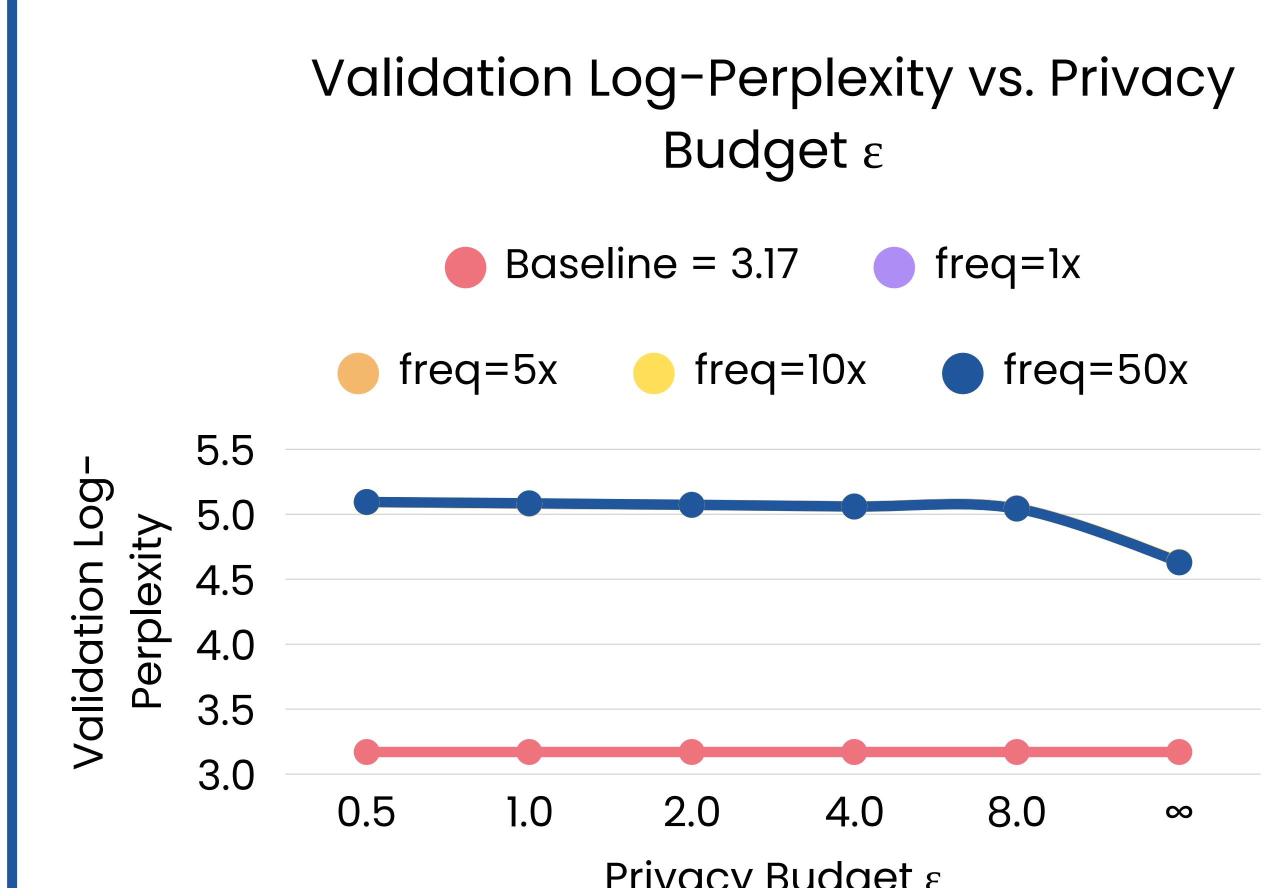
$$exposure = \log_2\left(900000\right) - \log_2\left(rank\right)$$

Methodology

1. Load the pretrained GPT-2 small model.
2. Select a mental health dataset from Hugging Face, insert canaries with varied frequencies.
3. Fine-tune the GPT-2 small model using next token prediction with varied privacy budgets ε.
4. Input the canary format into the fine-tuned model and check the rank of the canary in the output distribution.
5. Measure the log-perplexity on the validation set for model performance.

Results





Conclusion

1. Even at ε = 8, DP effectively reduces unintentional memorization of the canaries.
2. The higher the privacy budget ε, the lower the validation log-perplexity.
3. The number of times canaries are inserted does not affect the validation log-perplexity.
4. The freq=10x elevation is a base-rate artifact confirmed by ablation tests.